hence is the one that is running slow. That each of us thinks the other's clock is running slow may seem paradoxical, but it's not. At constant velocity, our clocks will continue to get farther apart and hence they don't allow for a direct, face-to-face comparison to determine which is "really" running slow. And all other indirect comparisons (for instance, we compare the times on our clocks by cell phone communication) occur with some elapsed time over some spatial separation, necessarily bringing into play the complications of different observers' notions of *now,* as in Chapters 3 and 5. I won't go through it here, but when these special relativistic complications are folded into the analysis, there is no contradiction between each of us declaring that the other's clock is running slow (see, e.g., E. Taylor and J. A. Wheeler, *Spacetime Physics*, for a complete, technical, but elementary discussion) . Where things appear to get more puzzling is if, for example, you slow down, stop, turn around, and head back toward me so that we can compare our clocks face to face, eliminating the complications of different notions of now. Upon our meeting, whose clock will be ahead of whose? This is the so-called twin paradox: if you and I are twins, when we meet again, will we be the same age, or will one of us look older? The answer is that my clock will be ahead of yours—if we are twins, I will look older. There are many ways to explain why, but the simplest to note is that when you change your velocity and experience an acceleration, the symmetry between our perspectives is lost—you can definitively claim that *you* were moving (since, for example, you *felt it*—or, using the discussion of Chapter 3, unlike mine, your journey through spacetime has not been along a straight line) and hence that your clock ran slow relative to mine. Less time elapsed for you than for me.

9. John Wheeler, among others, has suggested a possible central role for observers in a quantum universe, summed up in one of his famous aphorisms: "No elementary phenomenon is a phenomenon until it is an observed phenomenon." You can read more about Wheeler's fascinating life in physics in John Archibald Wheeler and Kenneth Ford, *Geons, Black Holes, and Quantum Foam: A Life in Physics* (New York: Norton, 1998). Roger Penrose has also studied the relation between quantum physics and the mind in his *The Emperor's New Mind,* and also in *Shadows of the Mind: A Search for the Missing Scienceof Consciousness* (Oxford: Oxford University Press, 1994).